

ST02 Fixed model paired BA contrasts

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| Stratum         | Paired contrast                | $\Delta$ BA (pp) | 95% paired CI (pp) | Conditional p | Holm p (six pairs) |
|-----------------|--------------------------------|------------------|--------------------|---------------|--------------------|
| OrganAMNIST / A | DenseNet-121 – EfficientNet-B0 | 2.838            | [2.689, 2.993]     | 0.0020        | 0.0117             |
| OrganAMNIST / A | DenseNet-121 – ResNet-18       | -0.440           | [-0.520, -0.362]   | 0.0020        | 0.0117             |
| OrganAMNIST / A | DenseNet-121 – ResNet-50       | -0.381           | [-0.464, -0.295]   | 0.0020        | 0.0117             |
| OrganAMNIST / A | EfficientNet-B0 – ResNet-18    | -3.278           | [-3.424, -3.138]   | 0.0020        | 0.0117             |
| OrganAMNIST / A | EfficientNet-B0 – ResNet-50    | -3.220           | [-3.352, -3.092]   | 0.0020        | 0.0117             |
| OrganAMNIST / A | ResNet-18 – ResNet-50          | 0.059            | [-0.010, 0.130]    | 0.0879        | 0.0879             |
| OrganAMNIST / B | DenseNet-121 – EfficientNet-B0 | 2.928            | [2.799, 3.052]     | 0.0020        | 0.0117             |
| OrganAMNIST / B | DenseNet-121 – ResNet-18       | -0.394           | [-0.463, -0.327]   | 0.0020        | 0.0117             |
| OrganAMNIST / B | DenseNet-121 – ResNet-50       | -0.411           | [-0.485, -0.337]   | 0.0020        | 0.0117             |
| OrganAMNIST / B | EfficientNet-B0 – ResNet-18    | -3.322           | [-3.425, -3.220]   | 0.0020        | 0.0117             |
| OrganAMNIST / B | EfficientNet-B0 – ResNet-50    | -3.339           | [-3.461, -3.226]   | 0.0020        | 0.0117             |
| OrganAMNIST / B | ResNet-18 – ResNet-50          | -0.016           | [-0.060, 0.025]    | 0.3672        | 0.3672             |

Six alphabetically oriented fixed-model contrasts per primary stratum. Intervals retain the existing paired split-then-seed procedure. Exact sign-flip p-values use split-average contrasts under conditional joint sign symmetry; Holm adjustment is within each six-comparison stratum. These are separate from the context-wise top-two order statistic.

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| Stratum      | Paired contrast                | $\Delta$ BA (pp) | 95% paired CI (pp) | Conditional p | Holm p (six pairs) |
|--------------|--------------------------------|------------------|--------------------|---------------|--------------------|
| SIPaKMeD / A | EfficientNet-B0 – ResNet-50    | -1.739           | [-2.137, -1.317]   | 0.0020        | 0.0117             |
| SIPaKMeD / A | EfficientNet-B0 – ResNet-18    | -1.968           | [-2.514, -1.345]   | 0.0020        | 0.0117             |
| SIPaKMeD / A | ResNet-18 – ResNet-50          | 0.229            | [-0.168, 0.635]    | 0.2578        | 0.2578             |
| SIPaKMeD / A | DenseNet-121 – ResNet-18       | -0.888           | [-1.373, -0.376]   | 0.0098        | 0.0391             |
| SIPaKMeD / A | DenseNet-121 – EfficientNet-B0 | 1.081            | [0.460, 1.674]     | 0.0098        | 0.0391             |
| SIPaKMeD / A | DenseNet-121 – ResNet-50       | -0.659           | [-1.015, -0.262]   | 0.0098        | 0.0391             |
| SIPaKMeD / B | DenseNet-121 – EfficientNet-B0 | 1.349            | [0.734, 1.956]     | 0.0078        | 0.0156             |
| SIPaKMeD / B | DenseNet-121 – ResNet-18       | -0.748           | [-1.073, -0.415]   | 0.0020        | 0.0117             |
| SIPaKMeD / B | DenseNet-121 – ResNet-50       | -0.771           | [-1.151, -0.366]   | 0.0039        | 0.0117             |
| SIPaKMeD / B | EfficientNet-B0 – ResNet-18    | -2.097           | [-2.600, -1.581]   | 0.0020        | 0.0117             |
| SIPaKMeD / B | EfficientNet-B0 – ResNet-50    | -2.121           | [-2.544, -1.718]   | 0.0020        | 0.0117             |
| SIPaKMeD / B | ResNet-18 – ResNet-50          | -0.023           | [-0.421, 0.368]    | 0.9023        | 0.9023             |

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| Stratum      | Paired contrast                | $\Delta$ BA (pp) | 95% paired CI (pp) | Conditional p | Holm p (six pairs) |
|--------------|--------------------------------|------------------|--------------------|---------------|--------------------|
| ISIC2019 / A | ResNet-18 – ResNet-50          | -4.256           | [-5.198, -3.234]   | 0.0020        | 0.0117             |
| ISIC2019 / A | EfficientNet-B0 – ResNet-50    | -5.891           | [-7.448, -4.262]   | 0.0020        | 0.0117             |
| ISIC2019 / A | EfficientNet-B0 – ResNet-18    | -1.635           | [-3.041, -0.197]   | 0.0391        | 0.1172             |
| ISIC2019 / A | DenseNet-121 – ResNet-18       | -0.326           | [-1.433, 0.723]    | 0.3301        | 0.3301             |
| ISIC2019 / A | DenseNet-121 – EfficientNet-B0 | 1.309            | [0.060, 2.515]     | 0.0684        | 0.1367             |
| ISIC2019 / A | DenseNet-121 – ResNet-50       | -4.581           | [-5.860, -3.362]   | 0.0020        | 0.0117             |
| MURA / A     | EfficientNet-B0 – ResNet-50    | -1.807           | [-2.455, -1.090]   | 0.0039        | 0.0156             |
| MURA / A     | DenseNet-121 – EfficientNet-B0 | 1.572            | [1.130, 1.964]     | 0.0020        | 0.0117             |
| MURA / A     | DenseNet-121 – ResNet-18       | 1.102            | [0.590, 1.698]     | 0.0020        | 0.0117             |
| MURA / A     | DenseNet-121 – ResNet-50       | -0.235           | [-0.818, 0.355]    | 0.4102        | 0.4102             |
| MURA / A     | EfficientNet-B0 – ResNet-18    | -0.470           | [-1.087, 0.226]    | 0.1133        | 0.2266             |
| MURA / A     | ResNet-18 – ResNet-50          | -1.337           | [-1.949, -0.742]   | 0.0039        | 0.0156             |

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